EUROPEAN
ADVISORY BOARD ON CAT DISEASES
ABCD

Feline leukaemia virus infection

FACT SHEET

What is feline leukaemia virus infection?

- Feline leukaemia virus (FeLV) infection can cause anaemia, immunosuppression, lymphoma and/or other conditions.
- It affects cats worldwide. The prevalence of infection in Europe varies; it is low in the North (≤1%), but higher in Southern countries, exceeding 20% in specific areas.

Transmission and pathogenesis

- Transmission occurs through viral shedding, mainly in saliva (but also via faeces etc.), through grooming, biting and blood transfusion.
- The virus does not remain infectious for long outside the host and is readily destroyed by disinfectants, soap, heating and drying.
- However, the virus can remain infectious longer under moist conditions, e.g. in faeces, contaminated needles, or in refrigerated blood for transfusions.
- FeLV infection usually starts in the oropharynx, which then can lead to a primary viraemia; infection can then potentially reach the bone marrow, leading to a secondary viraemia.
- In FeLV-infected cats, viral RNA is reverse-transcribed into viral DNA, which is usually incorporated into the host (cat) cell genome (as a provirus).
- Four outcomes of FeLV infection are defined:
  - **Progressive infection**: The viraemia persists lifelong. These cats shed FeLV permanently and are continuously both FeLV antigen test and proviral PCR positive. Most of them will develop FeLV-associated diseases, but with proper care and an indoor lifestyle, some can remain healthy for years.
  - **Regressive infection**: The viraemia is transient (or may not be detectable), but FeLV proviral DNA remains integrated into the DNA of host cells. Such cats are proviral PCR positive, antigen test negative, usually healthy and do not shed FeLV. Stress and/or corticosteroids can reactivate viraemia.
  - **Abortive infection**: No viraemia occurs, cats do not shed FeLV and remain healthy. Cats are both antigen and proviral PCR negative. Anti-FeLV antibodies are present in serum, but reliable detection of these antibodies is not commonly available, except in the UK.
  - **Focal infection**: Likely rare; the cat's immune system keeps viral replication sequestered in certain tissues. The cat is healthy and test results can be discordant.
- Young kittens are especially susceptible to FeLV infection and are more likely to develop progressive infection.

Clinical signs

- The most common signs of progressively FeLV infected cats are:
  - Anaemia (mainly non-regenerative)
  - Immune suppression (predisposition to other infections)
  - Some lymphomas (anterior mediastinal lymphoma, spinal)
- Less common signs:
  - Other lymphomas (e.g. gastrointestinal, renal, cutaneous, ocular)
  - Other neoplasms
  - Immune-mediated diseases (e.g. uveitis)
  - Reproductive disorders in progressively infected queens (foetal resorption, abortion, neonatal death and fading kittens)
  - Peripheral neuropathies (e.g. anisocoria, mydriasis, Horner's syndrome, abnormal vocalisation, hyperaesthesia, paresis, paralysis).

Please see the **ABCD FeLV diagnostic tool** for more detailed information on diagnostic tests

Diagnosis

- Routine testing focuses mainly on the detection of progressively infected cats, as, in contrast to the other outcomes of infection, these cats permanently shed FeLV and usually develop disease.
- Tests are used that detect:
  - **FeLV p27 antigen** (including in-house tests): a positive test result indicates the presence of FeLV antigenaemia, usually consistent with viraemia.
    - This test is persistently positive in progressively infected cats and can be transiently positive in regressively infected cats that have a transient viraemia (but not all regressively infected cats have a detectable viraemia).
    - Antigen-positive cats without clinical signs should be retested after 6-12 weeks to help determine if the viraemia is permanent (i.e. in a progressively infected cat).

    ***Note**: As the prevalence of FeLV infection decreases in a given cat population, the percentage of positive antigen test results that are false positives increases (i.e. the positive predictive value of the test result will decrease). Thus, a positive test result in a healthy cat should always be confirmed, preferably by proviral PCR.*
  - **FeLV proviral DNA**: PCR remains positive in both progressively and regressively infected cats.
  - **FeLV RNA**: a positive RT-PCR result from saliva is a reliable indicator of viraemia. The test becomes positive very early following infection, even before p27 antigen testing becomes positive.
- Testing for FeLV antibody in-house for the p15E envelope transmembrane protein is available in some countries; however, so far, the test does not reliably detect cats with abortive FeLV infection under field conditions. Assays for FeLV viral neutralising antibodies, which do reflect immunity to FeLV, are not widely available (except in the UK).

The ABCD is an independent panel of experts in feline health supported by Boehringer Ingelheim (founding sponsor), MSD Animal Health, Vetoquinol and Virbac. November 2024.

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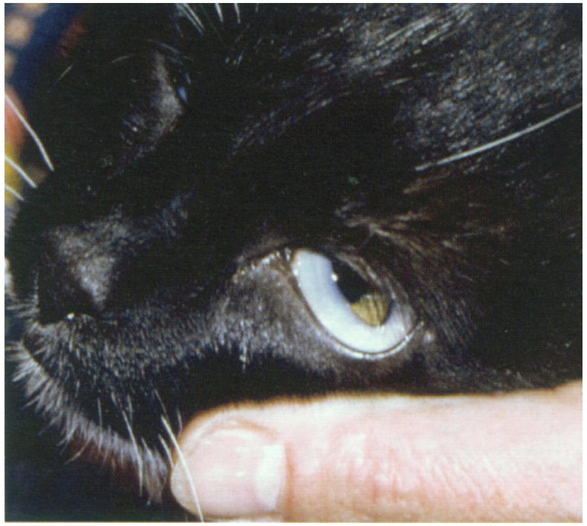
Infection management

- No treatments exist that can reverse progressive viraemia.
- Regressively infected cats are usually healthy and do not need antiviral drugs.
- Healthy progressively infected cats do not need antiviral drugs but should remain indoors, should not be co-housed with FeLV-negative cats and should receive clinical check-ups every 6 months.
- In cats showing FeLV-associated disease; good supportive and preventative care is essential. Secondary infections and neoplasms (e.g. lymphoma) should be treated promptly. Feline interferon omega, or zidovudine, might reduce clinical signs and extend survival time, but with zidovudine side effects can occur.
- Immunosuppressive drugs (e.g. high dosage corticosteroids) should be avoided in infected cats, if possible.
- Blood donors should be tested by PCR for FeLV provirus, as blood from regressively infected cats can transmit the infection.



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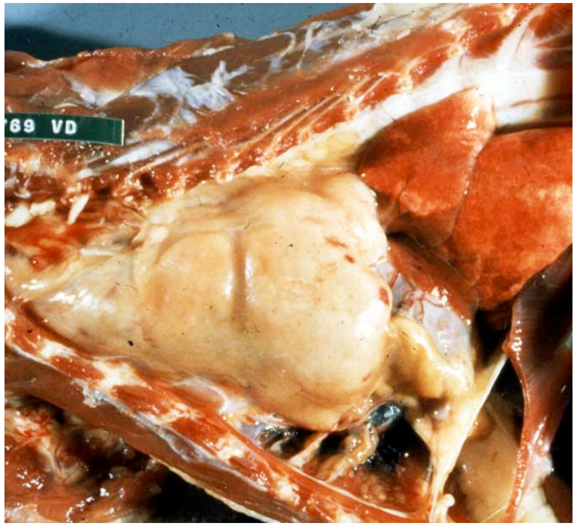
- *Anaemia in a cat with progressive FeLV infection. Most progressively infected cats develop clinical signs typical for FeLV infection.*



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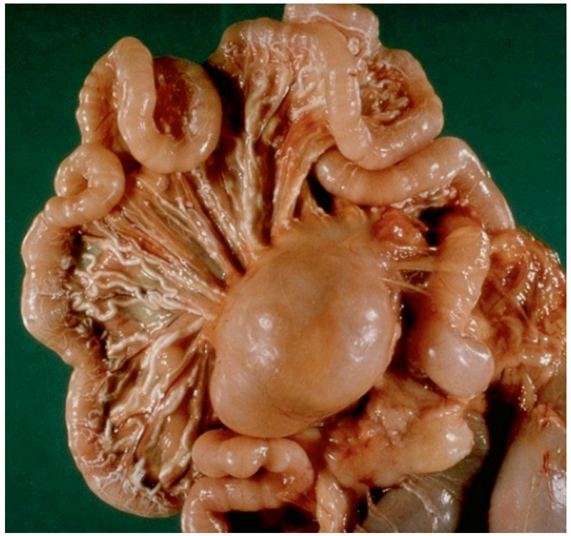
- *Anterior mediastinal lymphoma in the thorax of an FeLV-infected cat.*

Vaccination recommendations

- Before vaccination, all cats of uncertain FeLV status should be tested for p27 antigen, and preferably also for FeLV provirus.
- Progressively infected cats should not be vaccinated against FeLV as vaccination will be of no benefit.
- Regressively infected cats do not need to be vaccinated against FeLV as they already have acquired immunity.
- FeLV vaccines should be given to all cats that have outdoor access in areas where FeLV may be prevalent or are in contact with cats of unknown FeLV status.
- Kittens should be vaccinated at 8-9 weeks of age, with a second dose at 12 weeks, followed by revaccination one year later.
- FeLV boosters should be given annually until the age of 3 years. In older cats, a booster immunisation every 2-3 years seems to be sufficient.
- Vaccination of FeLV antigen-positive healthy cats with core vaccines for feline herpesvirus (FHV), calicivirus (FCV) and panleukopenia viruses (FPV) should be maintained. Indeed, more frequent boosters should be considered since protection might be incomplete and/or short-lived. Alternatively, for FPV, serum antibodies can be measured to determine if FPV vaccination is necessary.



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- *Mesenteric lymphoma associated with FeLV infection.*